Pauta

a) a.R/ $\vec{\mu} = I\vec{A} \rightarrow \vec{\mu} = (5)(10)(\hat{\imath} - \hat{\jmath}) \rightarrow \boxed{\vec{\mu} = 50(\hat{\imath} - \hat{\jmath})\ [A \cdot m^2]}$ (3ptos)

b.R/ $\vec{\tau} = \vec{\mu} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 50 & -50 & 0 \\ 0 & 5 & -5 \end{vmatrix}$

$$= (250 - 0)\hat{\imath} - (-250 - 0)\hat{\jmath} + (250 - 0)\hat{k}$$

Luego: $\vec{\tau} = \vec{\mu} \times \vec{B} = 250(\hat{\imath} + \hat{\jmath} + \hat{k})\ [N \cdot m]$ (5ptos)

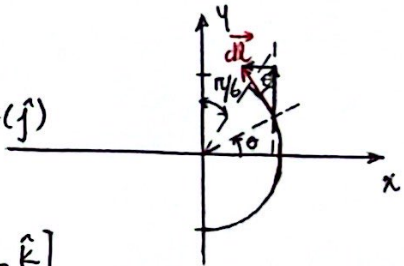
b) a.R/ Segmento recto

$\vec{F}_B = I_0 \int \vec{dl} \times \vec{B}$; $\vec{dl} = dx\,\hat{\imath}$; $\vec{B} = B_0 \hat{k}$

$$\vec{F}_B = I_0 \int_{-L}^{0} dx\, \overbrace{\hat{\imath} \times B_0 \hat{k}}^{(-\hat{\jmath})} = -I_0 B_0 \int_{-L}^{0} dx (\hat{\jmath}) = -I_0 B_0 \left(x \Big|_{-L}^{0}\right) \hat{\jmath}$$

$\boxed{\vec{F}_B = -I_0 B_0 L (\hat{\jmath})\ [N]}$ (5ptos)

b.R/ Segmento curvo



$\vec{F}_B = I_0 \int \vec{dl} \times \vec{B}$; $\vec{dl} = R\,\text{sen}\theta\, d\theta(-\hat{\imath}) + R\cos\theta\, d\theta(\hat{\jmath})$; $\vec{B} = -B_0 \hat{k}$

$$\vec{F}_B = I_0 \int \left[R\,\text{sen}\theta\, d\theta(-\hat{\imath}) + R\cos\theta\, d\theta(\hat{\jmath})\right] \times \left[-B_0 \hat{k}\right]$$

$$\vec{F}_B = I_0 R B_0 \int \left[\text{sen}\theta\, d\theta\, \overbrace{(\hat{\imath}) - \cos\theta\, d\theta\, (\hat{\jmath})] \times \hat{k}}^{(-\hat{\jmath})}$$

$(\hat{\imath})$

$$\vec{F}_B = I_0 R B_0 \left[-\int_{-\pi/2}^{\pi/3} \text{sen}\theta\, d\theta\, \hat{\jmath} - \int_{-\pi/2}^{\pi/3} \cos\theta\, d\theta\, \hat{\imath}\right] = I_0 R B_0 \left[-\left(-\cos\theta \Big|_{-\pi/2}^{\pi/3}\right)\hat{\jmath} - \left(\text{sen}\theta \Big|_{-\pi/2}^{\pi/3}\right)\hat{\imath}\right]$$

$$\vec{F}_B = I_0 R B_0 \left\{\left[\underbrace{\cos \pi/3}_{1/2} - \underbrace{\cos(-\pi/2)}_{0}\right]\hat{\jmath} - \left[\underbrace{\text{sen}\,\pi/3}_{\sqrt{3}/2} - \underbrace{\text{sen}(-\pi/2)}_{-1}\right]\hat{\imath}\right\}$$

$\boxed{\vec{F}_B = I_0 R B_0 \left[-\left(1 + \frac{\sqrt{3}}{2}\right)\hat{\imath} + \frac{1}{2}\hat{\jmath}\right]\ [N]}$ (5ptos)

c.R/ $\vec{F}_{B\,Total} = \vec{F}_{B\,Recto} + \vec{F}_{B\,curvo}$

$\vec{F}_{B\,Total} = I_0 B_0 \left[-\left(1 + \frac{\sqrt{3}}{2}\right)R\hat{\imath} + \left(\frac{1}{2}R - L\right)\hat{\jmath}\right]\ [N]$ (2ptos)